

Joe Hsu • Yingliang Hsu

AI / Computer Vision / Automation Engineer

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PROFESSIONAL SUMMARY

Automation Technology M.S. (NTUT) specializing in deep learning, computer vision, and industrial AOI delivery. Experienced from model training and inference optimization to cross-functional validation—turning algorithms into measurable, deployable systems.

KEY RESULTS

94.44% AOI Recall 7% PUE Reduction 15% LED Volume Cut 10% Brightness Gain

TECHNICAL SKILLS

Languages	Python • C/C++ • C# • JavaScript • VBA
AI / CV	PyTorch • TensorFlow • OpenCV • Scikit-learn • NumPy • Pandas
Deployment	ONNX • TensorRT • OpenVINO • Docker • Linux
Engineering	Git • GitHub Actions • Pytest • CI Workflows

PROFESSIONAL EXPERIENCE

Software Engineer 2023/07 – Present

Yifong Enterprise Co., Ltd. • New Taipei City

- Built Python automation and cross-platform data translation for API schema mapping, SKU attribute processing, and unstructured data conversion
- Designed rule-based logic using ROI, traffic cost, and product conditions to support automated filtering, ranking, and resource allocation
- Improved unattended script reliability with Git version control, exception handling, and log tracking

Automotive LED R&D Engineer (Alternative Service) 2022/01 – 2023/06

Everlight Electronics Co., Ltd. • New Taipei City

- Implemented top-contact thermal path and new die design, achieving 15% volume reduction and 10% brightness gain while completing automotive reliability validation
- Managed APQP/AEC-Q gates and coordinated QA/process partners to shorten validation cycles

CNC Technician (Intern) 2016/08 – 2018/07

Chicony Power Technology Co., Ltd. • New Taipei City

- Developed deep-bore fixtures to suppress vibration and drift, cutting jig cost by 20%
- Partnered with R&D on DFM reviews and tool-path optimization, lowering error rate by 30%

SELECTED PROJECTS

PCB Hybrid Defect Detection System (Featured)

C++ • Python • OpenCV • OpenVINO • Deep Learning • Link

- Combined rule-based ROI localization with DL classification for low-latency, high-accuracy inspection on industrial IPCs without high-end GPUs
- CV engine localized ROI in <10ms, cutting ~80% AI inference cost; OpenVINO INT8 optimized Intel CPU inference
- DL engine classifies 6 defect types; Pytest + CI keep algorithm iterations stable

Data Center Cooling Optimization

Python • DE-MRBFNN • NSGA3

- Built DE-MRBFNN + NSGA3 energy model for an air-cooled modular data center, cutting PUE by 7% in experiments
- Reduced CFD simulation runtime from hours to seconds; published in a peer-reviewed symposium

Anime4K C# Image Enhancer

C# • .NET 9 • WinForms • Link

- Implemented Anime4K core algorithm with WinForms live tuning for anime-style edge and texture enhancement
- Used LockBits for CPU-efficient memory access; real-time Scale/Strength/iteration controls

EDUCATION

M.S., Automation Technology 2019/08 – 2021/09

National Taipei University of Technology

- Multi-objective data-center energy model (DE-MRBFNN + NSGA3); 7% PUE reduction; symposium paper

B.S., Mechanical Engineering 2015/09 – 2019/06

National Taipei University of Technology

- Mechanical/manufacturing foundation supporting later electrical, optical, thermal, and system-integration work